

第二章 函数的导数(Derivative)

2.1 导数的定义(The Definition of Derivative)



Multiple Choice Questions

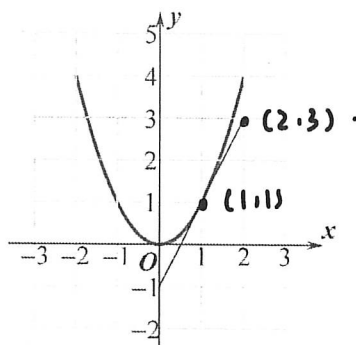
1. Which of the following values is the average rate of change of $f(x) = \sqrt{x+2}$ over the interval $[-1, 2]$?

- (A) -4 (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) 3

2. Let $f(x) = 4 - 3x$. Which of the following is equal to $f'(0)$?

- (A) 4 (B) -4 (C) 3 (D) -3

3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1) =$ ().



- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

4. Let $f(x) = kx^3 - 2x$, ($k \neq 0$). The average rate of $f(x)$ over the interval $[-2, -1]$ is 3, find k .

- (A) $\frac{6}{7}$ (B) $\frac{1}{7}$ (C) $-\frac{1}{7}$ (D) $-\frac{5}{7}$

5. **Calculator** Let $y = \tan^{-1} x - x^2 + 1$. Find $\frac{dy}{dx} \big|_{x=0.5} =$

- (A) -12.056 (B) 0 (C) 271.066 (D) 273.066

6. **Calculator** The graph of $y = \sqrt{x^2 + 1} - 3x$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

- (A) 0.354 (B) -0.001 (C) -2.666 (D) 0

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{2}{1}$$

$$69k = 3$$

$$\frac{2 - k + 8k - 4}{1}$$

$$7k = 3$$

$$7k = -3$$

$$k = -\frac{3}{7}$$

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$

$$\tan^{-1} x - x^2 + 1$$

$$-\tan^{-1} x + 2x$$

$$h$$

$$3x = \sqrt{x^2 + 1}$$

$$9x^2 = x^2 + 1$$

$$8x^2 = 1$$

$$x^2 = \frac{1}{8}$$

$$x = \frac{\sqrt{2}}{4}$$

7. The population of insects at time t days is modeled by the function P with first derivative $P'(t) = 0.1t^2 + 10t + 120$. Which of the following statements is true?

- (A) At time $t=10$, the population of insects is 230.
 (B) At time $t=10$, the population of insects is increasing at a rate of 230 insects per day.
 (C) At time $t=10$, the population of insects is decreasing at a rate of 230 insects per day.
 (D) At time $t=10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

8. If the line tangent to the graph of a function g at the point $(1, 4)$ passes through the point $(-2, -5)$, what is the slope of this line?

- (A) 3 (B) -3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

Free Response Questions

9. Let $f(x) = x^2 - 2x$

(a) Find the average rate of change of f over $[1, 3]$.

(b) Find $\frac{f(2+h) - f(2)}{h}$.

(c) Find the instantaneous rate of change of f at $x=2$.

$$(b): \frac{f(2+h) - f(2)}{h} = \frac{(2+h)^2 - 2(2+h) - 4 + 4}{h}$$

$$(c): \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{4h^2 + 4h - 4 + 4}{h} = \lim_{h \rightarrow 0} \frac{4h^2 + 4h}{h} = \lim_{h \rightarrow 0} (4h + 4) = 4$$

10. Use the definition of derivative to find the derivative of $f(x) = x^3$, then find $f'(2)$.

$$\frac{f(x+h) - f(x)}{h}$$

$$= \frac{(x+h)^3 - x^3}{h}$$

$$= \frac{(x^3 + 3x^2h + 3xh^2 + h^3) - x^3}{h}$$

$$= \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$$

$$= \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2$$

$$f'(x) = \lim_{h \rightarrow 0} (4x^2 + 2xh + h^2)$$

$$= 4x^2$$

$$\therefore f'(2) = 16$$